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AB
B

Test Code: MSQE Syllabus and Sample questions 1999

Syllabus for MEI (Mathematics)

Matrix Algebra : Matrices and Vectors, Matrix Operations, Determinants, non-singularity, Inversion, Cramer's rule.

Calculus : Limits, Continuity, Differentiation of functions of one or more variables, Product rule, Partial and total derivatives, Derivatives of implicit functions, unconstrained optimisation (first and second order conditions for extrema of functions of single variable, and several variables), Taylor series, Definite and indefinite integrals standard formulae, integration by parts and integration by substitution, Differential equations.

Constrained optimisation of functions of a single variable.

Theory of sequences and series.

Linear programming : Formulation, Statements of primal and dual problems, Graphical solutions.

Theory of polynomial equations (up to third degree).

Sample Questions for MEI (Mathematics)

Each question carries 4 marks.

1. The sum of the following series

$(n)(1) + (n-1)(3) + (n-2)(5) + \dots + (1)(2n-1)$

is

(A) $n^3 - \frac{n(n-1)(2n+5)}{6}$ (B) $n^3 - \frac{n(n-1)(5n-1)}{6}$

(C) $\frac{n(n+1)(2n+1)}{6}$ (D) none of these

2. If

$$A = \begin{pmatrix} 2 & 5 \\ 3 & 1 \end{pmatrix}$$

and

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

then the matrix $A^2 - 3A - 13I$ equals

(A) 0;

(B) I;

(C) $\begin{pmatrix} 15 \\ 30 \end{pmatrix}$;

(D) none of these.

[1]

3. If the inequality $\frac{mx^2+3x+4}{x^2+2x+2} < 5$ is satisfied for all real number x , then m satisfies the relation

(A) $m = 5$ (B) $m < 5$ (C) $5 < m < 7$ (D) $m < 5$ or $m > 7$

4. The number of solutions to

$$x + y = 2$$

$$y + z = 2$$

$$2x + 3y + z = 5$$

is

(A) infinite (B) zero (C) one (D) more than one but finite

5. Let

$$f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2 \sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$$

then $\lim_{x \rightarrow 0} \frac{f(x)}{x^3}$ is

(A) 0 (B) 1 (C) -1 (D) none of these

6. The value of the function

$$f(x) = \frac{1 - \cos 7(x - \pi)}{(x - \pi)} \text{ at } x = \pi \text{ so that } f \text{ is continuous at } \pi \text{ is}$$

(A) 0 (B) 2 (C) 7 (D) none of these

7. The real numbers x such that

$$x^2 + 4|x| - 4 = 0$$

are

(A) $\pm(\sqrt{8}-2)$; (B) $-2 \pm \sqrt{8}$;
(C) $2 \pm \sqrt{8}$; (D) -2 ± 8 and $2 \pm \sqrt{8}$.

8. The following matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & -1 \\ 4 & 1 & 0 & 1 \\ 8 & 1 & 0 & -1 \end{bmatrix}$$

- (A) is singular and the rank is 3
 (B) is singular and the rank is less than 3
 (C) is nonsingular and the determinant is -12
 (D) none of these.

9. The roots of the cubic equation $x^3 - 9x^2 + 23x - 15 = 0$ are in AP. The common difference between the roots is

- (A) $1/2$ (B) 2 (C) 3 (D) none of these.

10. If $\alpha_1, \alpha_2, \alpha_3, \dots$ are in GP, then the value of

$$\Delta = \begin{vmatrix} \log \alpha_n & \log \alpha_{n+1} & \log \alpha_{n+2} \\ \log \alpha_{n+3} & \log \alpha_{n+4} & \log \alpha_{n+5} \\ \log \alpha_{n+6} & \log \alpha_{n+7} & \log \alpha_{n+8} \end{vmatrix}$$

is

- (A) $\log(\alpha_n \alpha_{n+8}) - 2 \log \alpha_{n+4}$ (B) $\log(\alpha_1 \alpha_3) - 2 \log \alpha_2$
 (C) 0 (D) 1.

11. The function $f(x)$ defined by

$$f(x) = \begin{cases} x & \text{when } 0 < x < 1 \\ 2-x & \text{when } 1 \leq x < 2 \\ x - \frac{1}{2}x^2 & \text{when } x \geq 2, \end{cases}$$

is

- (A) continuous at $x = 2$ but not differentiable at $x = 1$
 (B) continuous at $x = 2$ and differentiable at $x = 1$
 (C) not continuous at $x = 2$ and not differentiable at $x = 1$
 (D) none of these.

12. $\lim_{x \rightarrow 0} \left[\frac{1}{x} - \frac{\log(1+x)}{x^2} \right]$ is

- (A) 1 (B) $1/2$ (C) does not exist (D) none of these.

13. If $|a| < 1$, $|b| < 1$, then the series

$$a(a+b) + a^2(a^2+b^2) + a^3(a^3+b^3) + \dots$$

- (A) converges to $\{a^2/(1-a^2) + b^2/(1-b^2)\}$
 (B) converges to $\{a^2/(1-a^2) + b^2/(1-ab)\}$
 (C) converges to $\{a^2/(1-a^2) + ab/(1-ab)\}$
 (D) none of the above is true.

14. If $a + \frac{1}{a} = 2 \cos \alpha$, then the value of $a^n + \frac{1}{a^n}$ is

- (A) $2^n \cos^n \alpha$ (B) $2^n \cos n\alpha$ (C) $2 \cos n\alpha$ (D) none of these.

15. If a, b and c are positive real numbers, then the least value of

$$(a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$$
 is

- (A) 2 (B) 3 (C) $\frac{10}{3}$ (D) none of these.

16. If

$$f(x) = \begin{cases} |x| - [x] - 1 & \text{if } -1 < x < 0 \\ |x| - [x] & \text{if } 0 \leq x < 1 \\ 1 & \text{otherwise,} \end{cases}$$

where $[x]$ is the greatest integer $\leq x$ and $|x|$ is the absolute value of x , then

- (A) $f(x)$ is a discontinuous function with points of discontinuity at only one point on the real line
 (B) $f(x)$ is a discontinuous function with points of discontinuity at more than one point on the real line
 (C) $f(x)$ is a continuous function but not differentiable at only one point on the real line
 (D) none of these.

17. If a primal linear programming problem has feasible solutions, but no optimal solution then its dual problem

- (A) may have an optimal solution
- (B) has no feasible solution
- (C) has feasible solutions, but no optimal solution
- (D) cannot be formulated.

18. For the following linear programming problem :

$$\text{maximise } 2x_1 + 3x_2$$

Subject to

$$2x_1 + x_2 \geq 2$$

$$x_1 + x_2 \leq 3$$

$$x_1 - x_2 \leq -1$$

$$x_1, x_2 \geq 0.$$

Which one of the following alternatives is true ?

- (A) the problem has an optimal solution at $(x_1 = 0, x_2 = 3)$
- (B) the problem has an optimal solution at $(x_1 = 1, x_2 = 2)$
- (C) the problem does not have an optimal solution
- (D) none of these

19. The value of $\int_{-1}^1 (x + x^2 + x^3)e^{-x^2/2} dx$ is

- (A) $2/\sqrt{e}$
- (B) $-2/\sqrt{e}$
- (C) $2\sqrt{e}$
- (D) $-2\sqrt{e}$

20. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$

- (A) converges absolutely
- (B) converges conditionally and the value is $\log_e 2$
- (C) does not converge
- (D) converges conditionally and the value is $\log_{10} 2$.

- (b) If he can borrow any amount at 5% per annum only to finance current rent consumption, how much should he consume each year?

6. Consider a small country facing a given world price for good x . There is a single domestic producer who is protected in the domestic market by tariff T per unit of output. In the world market, however, he has to sell at the competitive world price. Assuming that the domestic producer has an upward rising marginal cost and that he sells both in home and abroad, show that a rise in tariff can not decrease his exports. Under what conditions will exports remain constant?

7. A good Q is produced using only input x . Let P be the price of the good Q . The market for Q is supplied by 100 identical competitive firms each of which has the production function $q = x$. Each firm behaves as if the price of x (denote it by P) were constant. Derive the industry's supply curve in each of the following cases. Also comment on the slope of the industry's supply curve.

(a) Suppose the industry as a whole faces a horizontal supply curve for input x such that $p_x = \text{Rs. } 10$ at any level of supply.

(b) Suppose the industry as a whole faces a vertical supply curve for input x such that its supply is 100 units irrespective of the price of x .

8. Instruction for question numbers 8(a)–8(h). For each of the following questions four alternative answers are provided. Choose the answer that you consider to be the most appropriate for a question and write it down in your answer book.

(a) In a simple Keynesian model of the macro economy where mps is marginal propensity to save. If private investment depends positively on aggregate income, the multiplier is

- (A) less than $1/\text{mps}$
- (B) greater than $1/\text{mps}$
- (C) equal to $1/\text{mps}$
- (D) equal to $1/(1-\text{mps})$.

AB

21. The solution of the following differential equation

$$(1+x)y \, dx + (1-y)x \, dy = 0$$

is

- (A) $\log |x| + \log |y| - x + y = \text{constant}$
- (B) $\log |x| - \log |y| - x + y = \text{constant}$
- (C) $\log |x| + \log |y| + x - y = \text{constant}$
- (D) $\log |x| - \log |y| + x - y = \text{constant}$

Syllabus : for MEII (Economics)

Microeconomics : Consumer theory, Producer theory, Market forms, Welfare economics. **Macroeconomics :** National income accounting, simple model of income determination and multiplier; Banking, money supply and inflation. Applicants are expected to have familiarity of these topics at the level of Samuelson and Nordhaus (Economics), and Lipsey (Positive Economics).

Sample Question for MEII (Economics)

1. Murthy takes only idli and vara for dinner. His preference pattern is given by the utility function $u = f(\sqrt{q_1} + \sqrt{q_2})$, $f'(\cdot) > 0$, q_1, q_2 being the number of pieces of idli and vara, respectively.

At the food shop prices are Re. 1 and Rs. 2 per piece of idli and vara respectively. Fractional quantities are not sold,

- (a) One day Murthy ordered and ate 6 pieces of each item. Afterwards he introspected whether the order was optimal for him. Do you think his order was optimal? If not, what should have been the optimal order, given that he spends no more than in the above order?

- (b) Next day Murthy finds the shop has revised prices and charges Rs. 1.50 per piece for both items. Murthy now decides to have previous day's optimal satisfaction level and spend no more than what he did the day before. What should he order for?

- ✓ 2. Malati and her three sisters own a small farm. The production function of the farm is given by

Number working on Farm	Output (in rupees)
One sister	1500
Two sisters	2500
Three sisters	3300
Four sisters	4000

Assume that the sisters work equally hard and the output is equally divided between them. Also, they have the option of migrating to the nearby city where each can expect to earn Rs. 1110.

How many sisters will move to the city when each is interested in maximizing (a) her own income ; (b) total family income ?

3. A good is produced using only one variable input according to the production function $q = \sqrt{x}$. There are 100 perfectly competitive firms in the market. Derive the supply curve of the industry and discuss the nature of its slope in each of the following cases :

- (a) The industry faces a horizontal supply curve of the input at price Rs. 10.00.
 (b) The industry faces a vertical supply curve of the input at the quantity of 100 units.

4. An industry demand equation is given by $P = a - Q$ and the marginal cost curve is represented by the equation $MC = Q$ where P is price, Q is quantity and MC is marginal cost. If the industry is served by a monopolist as compared to perfectly competitive firms with the same marginal cost functions, calculate the deadweight loss.

5. A farmer finds that for every 100 bushels of corn that he sets aside this year, he can reap 110 bushels next year. He has now 1000 bushels. His utility depends on the quantity of corn he consumes this year and next, according to the function $W = C_0 C_1$, where C_0 is consumption this year and C_1 consumption next year. (Assume corn is perfectly divisible.)

- (a) If there is no capital market, how much should he consume this year and the next to maximise utility ?

(b) A farmer produces 100 units of rice. He keeps a part of his output for self-consumption and sells the rest in the market to buy vegetables. If the price of rice goes up, with the price of vegetables remaining unchanged, his budget line

- (A) shifts out with the same intercept on the rice-axis.
- (B) shifts out with the same intercept on the vegetable-axis.
- (C) does not change
- (D) shifts in with the same intercept on the rice-axis.

(c) A newly established firm sold Rs. 50,000 worth of its products and carried over a stock of Rs. 7000 of unsold products to the next year in the first year of its operation. In the same year it bought the product of other firms worth Rs. 30,000 for use as raw materials etc, of which it carried over a stock of unused products worth Rs. 2000 to the next year. The Value added in its production in the year was

- (A) Rs. 20,000
- (B) Rs. 25,000
- (C) Rs. 27,000
- (D) Rs. 29,000

check (d) Consider an economy where households have 100 units of labour and 1000 units of capital. To produce one unit of output only half unit of labour and four units of capital are needed. The average as well as the marginal propensity to consume is $\frac{1}{2}$ for every household. The amount of autonomous expenditure (in units of output) needed to sustain the maximum feasible level of production is

- (A) 100
- (B) 125
- (C) 150
- (D) 200

(e) A price taking firm pays a tax of t rupees per unit of output, and its production cost is $C(q, k) + K$, where q is the output, K denotes fixed cost, $C(q, k)$ denotes variable costs, and k is some parameter of the cost function. Suppose $\frac{\partial C}{\partial k} < 0$, i.e. the firm becomes more efficient if k increases, $\frac{\partial C}{\partial q} > 0$, and $\frac{\partial^2 C}{\partial q^2} > 0$. Which of the following statements is necessarily correct? If k increases (but t does not change)

- (A) the firm must pay less in taxes
- (B) the firm must pay more in taxes
- (C) the firm's profit after paying taxes increases
- (D) none of the above

- (f) A "Vickery Auction" is an auction in which each bidder submits a sealed bid (i.e. a bid that is revealed only after all the bids are submitted). The highest bidder wins the auction, but the winning bidder pays not what he bids, but the second highest bids. For example, if the highest bid was Rs. 100 and the second highest bid was Rs. 98, the winner (the person bidding Rs. 100) pays only Rs. 98.

Suppose you are participating in a "Vickery Auction" for an object (a painting) which is worth Rs. 20,000 to you. How much will you bid for the painting?

- (A) Rs. 19,600 (B) Rs. 19,998
☒ (C) Rs. 20,000 (D) Rs. 20,002

- (g) Suppose an individual lives for two periods. In period 1 this individual earns an income Y and saves an amount S . Savings are deposited in a bank and earn an interest of 10% per period. In period 2 the individual does not earn anything but consumes the entire savings and interest earned on it. If now the interest rate increases to 12%, this individual's saving will

- (A) increase (B) fall
(C) remain unchanged ☒ (D) be any of the above

- (h) In this question you are to choose one of (A)–(D) to fill in the two successive blank places in the following statement:

Consider the commodity and money market relationships in the standard macro model. While _____ interest rate affects investment and consumer spending (in the commodity market relationship), _____ interest rate affects the real demand for money.

- (A) real, real (B) real, nominal
(C) nominal, nominal (D) nominal, real